Milestone 5

Model Evaluation & Analysis

Hindi → Haryanvi → Speech Pipeline

Overview

- Text-to-Text: **LLaMA 3.1 (QLoRA)**
- Text-to-Speech: **Coqui VITS**
- End-to-End Dialect Pipeline

Evaluation Dataset

- Text Dataset: ~5000 samples
- Train: 4000 | Val: 500 | Test: 500
- TTS Dataset: ~5500 samples
- Filtered: ~2764 (~1.5 hours audio)

Preprocessing

- Duplicate & null removal
- Sentence length filtering
- Tokenization (LLM)
- Audio resampling (22050 Hz)
- LJSpeech format conversion

Evaluation Environment

- GPU: NVIDIA A100 (40GB)
- Python 3.12
- PyTorch 2.10
- Transformers 5.0
- BF16 Precision

Evaluation Metrics

- **Translation:** Exact Match, BLEU, chrF
- **TTS:** Mel Loss, KL Loss
- Generator & Discriminator Loss
- Duration Loss

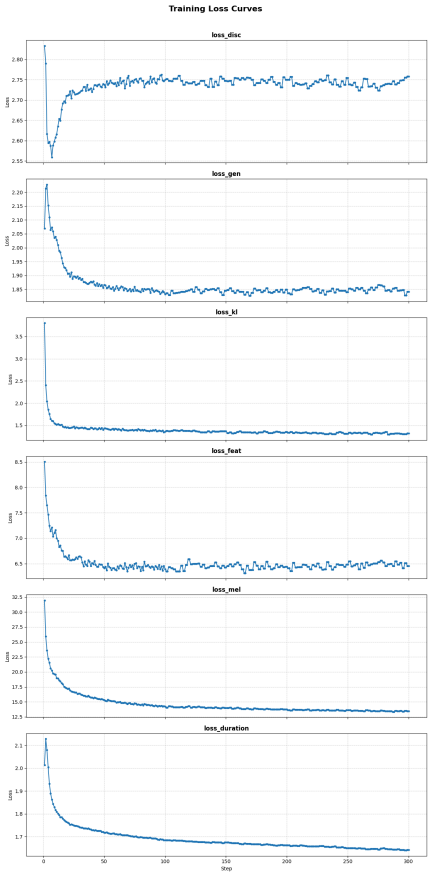
Best Model Results

Metric	Value
Eval Loss	0.00324
BLEU	71.76
chrF	77.65
Exact Match	50.42%

Experiment Comparison

Experiment	Eval Loss
baseline	0.00436
lower_lr_longer	**0.00324**
higher_rank	0.00427

TTS Training Curves



Qualitative Results

- ✓ तुम कहाँ जा रहे हो → तू कड़े जा रया सै
- ✓ मुझे पानी चाहिए → मन्ने पाणी चाहिये
- ❌ नहीं → कोनी sometimes missed

Error Analysis

- Proper nouns not handled
- Idiomatic expressions fail
- Complex sentences cause errors
- TTS distortion in long inputs

Limitations

- **Evaluation bug (same metrics across runs)**
- Small dataset
- No MOS evaluation

Conclusion

Strong translation + stable TTS pipeline

Promising for low-resource dialect AI